

WEEK 2: EXPLORATORY DATA ANALYSIS (EDA) AND VISUALIZATION FRAMEWORK DESIGN

AI-Based Pothole Detection and Road Condition Analysis Using Python

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Task	Week 2: EDA and Visualization Framework Design

1. Introduction and Purpose

Exploratory Data Analysis (EDA) examines data quality, distributions and relationships before model development. This report designs a proposed EDA and visualization framework for a pothole-detection system. It describes planned checks, charts and reporting methods. No real dataset has been analyzed and no empirical results are claimed.

2. Proposed Dataset and Data Dictionary

The future dataset will contain appropriately licensed road photographs and, where available, bounding-box annotations around potholes. The following illustrative metadata fields would support exploratory analysis:

Potential Field	Type	Purpose
image_id	Text	Unique image reference
image_width / image_height	Integer	Resolution and aspect-ratio checks
file_size_kb	Numeric	Identify unusually sized files
pothole_count	Integer	Annotations per image
bbox_width / bbox_height	Numeric	Object size distribution
bbox_area_ratio	Numeric	Object area divided by image area
lighting / weather	Category, if labeled	Inspect dataset diversity
source / split	Category	Check leakage between splits

3. EDA Questions and Exploration Methods

Approach	Planned Checks
Univariate	Image dimensions, file size, pothole count, box area and lighting frequencies.
Bivariate	Resolution versus pothole count; lighting versus box size; source versus annotation count.
Multivariate	Jointly compare image source, lighting, resolution and pothole frequency.
Image-level review	Inspect representative images, difficult lighting and potential annotation errors.

4. Data Quality and Cleaning Strategy

- Verify file readability, image format, dimensions and annotation schema.
- Find exact duplicates using file hashes; manually review possible near-duplicates.
- Check missing image files, metadata and labels separately; no-pothole images can be valid negatives.
- Validate bounding-box coordinates and flag suspiciously small or large boxes.
- Investigate outliers without automatically removing rare but valid road conditions.
- Group related frames before train/validation/test splitting to prevent leakage.
- Record all cleaning decisions in a data-quality log.

5. Proposed Visualization Strategies

Chart / Visual	Purpose	Python Tool
Bar chart	Images by source, lighting or category	Seaborn / Plotly
Histogram	Image dimensions, pothole counts, box area ratios	Matplotlib
Box plot	Compare box sizes across sources or lighting	Seaborn
Scatter plot	Resolution versus object size or count	Matplotlib / Plotly
Correlation heatmap	Relationships among meaningful numeric features	Seaborn
Image contact sheet	Visually review images and bounding-box overlays	OpenCV / Matplotlib
Missingness matrix	Metadata completeness	Pandas / Seaborn

6. Planned EDA Workflow

Inventory images and metadata → Validate files and labels → Generate summary statistics → Review duplicates, missing values and outliers → Create univariate plots → Compare selected variables → Inspect annotated sample images → Record findings and recommendations.

7. Python Tools and Libraries

- Pandas: metadata tables, missing-value checks, grouping and descriptive statistics.
- NumPy: numeric calculations, quantiles and bounding-box features.
- OpenCV / Pillow: image checks, preview resizing and annotation overlays.
- Matplotlib: static histograms, scatter plots and sample-image grids.
- Seaborn: category comparisons, box plots and correlation heatmaps.
- Plotly: optional interactive charts and filters.
- Jupyter Notebook: reproducible notes, analysis steps and charts.

8. Documentation and Reporting Plan

A later EDA report would include dataset provenance and licensing, dataset schema, data-quality summaries, descriptive statistics, properly labeled charts, representative image examples, limitations and recommendations for preprocessing. Actual observations will be distinguished from illustrative examples.

9. 35-Hour Project Timeline

Stage	Planned Work	Hours
EDA planning	Define questions and dataset schema	4
Data validation	Plan image and annotation checks	5
Cleaning strategy	Design duplicate, missingness and outlier checks	5
Exploration strategy	Plan univariate and bivariate analysis	6
Visualization design	Select charts and sample-image layouts	6
Tools and notebook	Choose libraries and notebook structure	3
Reporting	Design interpretation and reporting template	3
Review	Refine and document framework	3
TOTAL	Total planned effort	35

10. Expected Outcomes and Limitations

Expected outputs are an actionable EDA checklist, a proposed data dictionary, a visualization matrix, a reproducible workflow and a reporting structure. This is a planning document; real plots, statistical findings and dataset-specific conclusions require a future dataset.

11. Potential Challenges and Mitigation

Challenge	Mitigation
Unbalanced road conditions	Compare sources, weather and lighting where metadata exists.
Annotation inconsistency	Define labeling rules and review sample boxes.
Data leakage	Group related frames and near-duplicates before splitting.
Misleading correlations	Inspect distributions and meaningful data types.
Privacy and licensing	Use permitted images and avoid exposing faces or number plates.

12. Conclusion

This Week 2 framework outlines how a future pothole-image dataset would be explored, cleaned, visualized and documented using Python. It prepares the project for informed preprocessing and later object-detection model development without claiming that analysis has already been performed.

13. References / Planned Resources

- Pandas: <https://pandas.pydata.org/docs/>
- Matplotlib: <https://matplotlib.org/stable/>
- Seaborn: <https://seaborn.pydata.org/>
- Plotly: <https://plotly.com/python/>
- OpenCV: <https://docs.opencv.org/>